Question 1

$$J = \int_0^1 \sqrt{1-x^4}\,dx$$
$$K = \int_0^1 \sqrt{1+x^4}\,dx$$
$$L = \int_0^1 \sqrt{1-x^8}\,dx$$

Order the definite integrals above and 1 from smallest to largest. Justify!
The order is $J < L < 1 < K$.

- $K > 1$ because at this interval, $\sqrt{1+x^4} \in (1, \sqrt{2})$, so the area must exceed 1.
- $1 > L, J$ because at this interval, $\sqrt{1-x^n} \in (0, 1)$, so the area must be less than 1.
- L and J is a decreasing function as $\sqrt{1-x^n}' = -\frac{nx^{n-1}}{2\sqrt{1-x^n}} < 0$. We can simply take a point in the interval and compare the values of the functions to see that $L > J$: $\sqrt{1-0.5^8} => \sqrt{1-0.5^4}$. Therefore, the area of $n = 8$ for $\sqrt{1-x^n}$ is greater than $n = 4$, so $L > J$.

Question 2

Make a substitution and use the method of partial fractions to evaluate

$$\int \frac{\cos x}{\sin^2 x + \sin x} dx.$$

$$\int \frac{\cos x}{\sin^2 x + \sin x} dx = \int \frac{\cos x}{\sin x(\sin x + 1)} dx$$
$$\text{Let } u = \sin x \to du = \cos x\,dx$$
$$\int \frac{1}{u(u+1)} du = \int (\frac{1}{u} - \frac{1}{u+1}) du$$
$$= \ln|u| - \ln|u+1| + c$$
$$= \ln|\sin x| - \ln|\sin x + 1| + c$$

Question 3

Complete the square and use trigonometric substitution to evaluate

$$\int \frac{dt}{\sqrt{t^2 - 6t + 13}}.$$

$$\int \frac{1}{\sqrt{t^2-6t+13}}\,dt = \int \frac{1}{\sqrt{(t-3)^2+4}}\,dt$$
$$= \int \frac{1}{2\sqrt{(\frac{t-3}{2})^2+1}}\,dt$$

Let $\dfrac{t-3}{2} = \tan\theta \to dt = 2\sec^2\theta d\theta$

$$\int \frac{2\sec^2\theta}{2\sqrt{\tan^2\theta+1}}\,d\theta = \int \frac{\sec^2\theta}{\sqrt{\sec^2\theta}}\,d\theta$$
$$= \int \sec\theta\,d\theta$$
$$= \ln|\sec\theta+\tan\theta| + c$$

$\sec\theta = \sqrt{1+\tan^2\theta} \to$
$$= \ln|\frac{t-3}{2} + \sqrt{(\frac{t-3}{2})^2+1}| + c$$
$$= \ln|t-3+\sqrt{(t-3)^2+4}| - \ln 2 + c$$
$$= \ln|t-3+\sqrt{(t-3)^2+4}| + c$$

Question 4

Use Integration by Parts to prove the reduction formula

$$I_n = \int x^n \cos x\,dx = x^n \sin x + nx^{n-1}\cos x - n(n-1)I_{n-2}.$$

$$\int uv = u\int v - \int (u' \times \int v)$$

Let $u = x^n \to u' = nx^{n-1}$

Let $v = \cos x \to \int v = \sin x$

$$I_n = x^n \sin x - \int (nx^{n-1}\sin x)\,dx$$

$$\int (nx^{n-1}\sin x)\,dx = -nx^{n-1}\cos x + n(n-1)\int x^{n-2}\cos x\,dx$$

$\because I_{n-2} = \int x^{n-2}\cos x\,dx \to$
$$= -nx^{n-1}\cos x + n(n-1)I_{n-2}$$

$$\therefore I_n = x^n \sin x - (-nx^{n-1}\cos x + n(n-1)I_{n-2})$$
$$= x^n \sin x + nx^{n-1}\cos x - n(n-1)I_{n-2}$$

Question 5

Question 5a

Use the Quotient Rule for differentiation and the Fundamental Theorem of Calculus to find a rule to integrate quotients.

$$
\begin{aligned}
(\frac{a}{b})' &= \frac{ba' - ab'}{b^2} \\
(\frac{a}{b})' &= \frac{a'}{b} - \frac{ab'}{b^2} \\
\frac{a'}{b} &= \frac{ab'}{b^2} + (\frac{a}{b})' \\
\int \frac{a'}{b} &= \int \frac{ab'}{b^2} + \frac{a}{b}
\end{aligned}
$$

Question 5b

Use your rule to integrate quotients to evaluate the following, and verify the correctness

$$\int \frac{\sin\left(x^{-\frac{1}{2}}\right)}{x^2} dx$$

Note that we are looking for a' consisting $\sin(x^{-\frac{1}{2}})$:

$$\text{Let } a = 2\cos(x^{-\frac{1}{2}}) \to a' = x^{-\frac{3}{2}}\sin(x^{-\frac{1}{2}})$$

$$\text{Let } b = x^{\frac{1}{2}} \to b' = \frac{1}{2}x^{-\frac{1}{2}}$$

$$
\begin{aligned}
\int \frac{a'}{b} &= \int \frac{ab'}{b^2} + \frac{a}{b} \\
\int \frac{\sin\left(x^{-\frac{1}{2}}\right)}{x^{(\frac{1}{2}+\frac{3}{2})}}\, dx &= \int \frac{2\cos(x^{-\frac{1}{2}})\frac{1}{2}x^{-\frac{1}{2}}}{x}\, dx + \frac{2\cos(x^{-\frac{1}{2}})}{x^{1/2}} \\
&= \int x^{-\frac{3}{2}}\cos(x^{-\frac{1}{2}})\, dx + 2\frac{\cos(x^{-1/2})}{x^{1/2}} \\
\because (2\cos(x^{-\frac{1}{2}}))' = x^{-\frac{3}{2}}\sin(x^{-\frac{1}{2}}) \to -2\sin(x^{-\frac{1}{2}}) &= \int x^{-\frac{3}{2}}\cos(x^{-\frac{1}{2}})\, dx \\
\therefore &= -2\sin(x^{-\frac{1}{2}}) + 2\frac{\cos(x^{-\frac{1}{2}})}{x^{\frac{1}{2}}} + c
\end{aligned}
$$

Verification:

$$
\begin{aligned}
Letu = x^{-\frac{1}{2}} \to du &= -\frac{1}{2}x^{-\frac{3}{2}}\ dx \\
\int \frac{\sin\left(x^{-\frac{1}{2}}\right)}{x^2} dx &= \int x^{-\frac{3}{2}} x^{-\frac{1}{2}} (-\frac{1}{2})(-2) \sin x^{-\frac{1}{2}}\ dx \\
&= \int -2u \sin u\ du \\
&= -2(-u\cos u - \int -\cos u\ du) + c \\
&= 2x^{-\frac{1}{2}} \cos x^{-\frac{1}{2}} - 2\sin x^{-\frac{1}{2}} + c \\
&= -2\sin(x^{-\frac{1}{2}}) + 2\frac{\cos(x^{-1/2})}{x^{1/2}} + c
\end{aligned}
$$

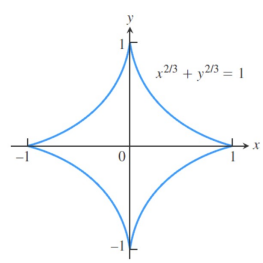
Question 6

A Bundt cake, well known for having a ringed shape, is formed by revolving around the y-axis the region bounded by the graph of $y = sin(x^2 - 1)$ and the x-axis over the interval $1 \leq x \leq \sqrt{1+\pi}$. Find the volume of the cake

$$
\begin{aligned}
V &= \pi \int_a^b 2(x+r)R(x)\ dx \\
r = 0 \to &= \pi \int_1^{\sqrt{1+\pi}} 2(x)(\sin(x^2-1))\ dx \\
(-\cos(x^2-1))' = 2x\sin(x^2-1) \to &= \pi[-\cos(x^2-1)]_1^{\sqrt{1+\pi}} \\
&= \pi(-\cos(1+\pi-1) + \cos(1-1)) \\
&= \pi(-\cos(\pi)+1) \\
&= \pi(1+1) \\
&= 2\pi
\end{aligned}
$$

Question 7

Find the perimeter of the shape formed by $x^{2/3} + y^{2/3} = 1$:



We can parametize the equation for easier calculation:

$$\begin{aligned}
x^{2/3} + y^{2/3} &= 1 \\
\sin^2 t + \cos^2 t &= 1 \\
x &= \sin^3 t \\
y &= \cos^3 t \\
x' &= 3\sin^2 t \cos t \\
y' &= -3\cos^2 t \sin t
\end{aligned}$$

To find the length of a side, we are interested in $x \in [0, 1] \to t \in [0, \pi/2]$:

$$\begin{aligned}
L &= \int_0^{\pi/2} \sqrt{x'^2 + y'^2}\, dt \\
&= \int_0^{\pi/2} \sqrt{9\sin^4 t \cos^2 t + 9\cos^4 t \sin^2 t}\, dt \\
&= 3\int_0^{\pi/2} \sin t \cos t\, dt \\
&= \frac{3}{2}\int_0^{\pi/2} \sin 2t\, dt \\
&= \frac{3}{2}[\frac{-\cos 2t}{2}]_0^{\pi/2} \\
&= \frac{3}{2}[\frac{-\cos \pi}{2} + \frac{\cos 0}{2}] \\
&= \frac{3}{2} \\
\therefore P &= 4 \times \frac{3}{2} = 6
\end{aligned}$$

Question 8

Use the integral formula for arc length to show that the shortest distance between two points is a line:

The arc length formula $L = \int_a^b \sqrt{1 + f'^2(x)}\, dx$ is simply the submation of the distance between two points. The horizontal distance between two points is dx, and the vertical distance is $dy = f'(x)dx$. Hence, the distance is given by the Pythagorean theorem, $\sqrt{1 + f'^2(x)} \times dx$.

There we can see how the arc length formula is defined as the sum of the **straight-line distance between points** in the interval $[a, b]$. Therefore, the shortest distance between two points is a line.